

Performance issue with particles causing lag

MC-124170 **Particles**

Status	Resolved / Fixed
Affected	Minecraft 18w02a, Minecraft 18w03b, Minecraft 18w05a, Minecraft 18w06a, Minecraft 18w07b, Minecraft 18w07c (+70 more)
Fixed in	Minecraft 1.14.3 Pre-Release 2
Reported	2018-01-11

STEPS TO REPRODUCE

1. Launch Minecraft 18w11a.
2. Create and enter a new Singleplayer Survival world with default settings.
3. Open Options -> Video Settings, set Render Distance to 2 chunks, and set the maximum FPS to 60.
4. Travel to the End and defeat the ender dragon.
5. Wait until the dragon shoots a fireball and an area-effect cloud particle is created.
6. Observe repeated lag spikes and reduced game performance after the area-effect cloud appears.
7. Summon a blaze using `/summon minecraft:blaze ~ ~ ~`.
8. Observe that the FPS decreases further when the blaze and its smoke particles are present.

OBSERVED BEHAVIOUR

Actual behavior

Particle rendering causes significant and inconsistent FPS degradation in Java Edition versions 17w50a-18w11a, compared with 1.12.2. The impact varies considerably by particle type: some particles have little or no effect, while others cause severe frame-rate drops even when only 100 particles are present.

Test conditions:

- CPU: Intel Core i5-4570 @ 3.20 GHz
- RAM: 8 GB
- GPU: GeForce GTX 1050 Ti OC Edition
- Render distance: 2 chunks
- FPS limit: 60
- Approximately 100 particles spawned continuously using an always-active repeating command block

Observed results include:

- ``angry_villager``: 33-35 FPS
- ``barrier``: 33-35 FPS
- ``bubble``: 60 FPS
- ``cloud``: 1-5 FPS
- ``crit``: 49-52 FPS
- ``damage_indicator``: 12-26 FPS
- ``dragon_breath``: approximately 55 FPS
- ``effect``: approximately 29 FPS
- ``enchant``: 60 FPS

Naturally generated particles also cause severe performance problems, particularly:

- ``dragon_breath``
- ``cloud``
- ``blockcrack``
- ``smoke`` generated by blazes and fire
- Rain particles

In Survival mode, after the Ender Dragon creates an `area\_effect\_cloud` from its fireball attack, the resulting particles cause repeated lag spikes. Summoning blazes also causes a noticeable FPS decrease.

Some particle-related performance issues were improved in snapshot 18w09a, but the problem remains for multiple particle types, including dragon breath and rain. Overall particle performance remains substantially worse than in 1.12.2, including on the reported hardware.

EXPECTED BEHAVIOUR

Particles should render smoothly without causing severe lag spikes or substantial FPS reductions, including area-effect-cloud, dragon breath, rain, cloud, block-crack, smoke, and other particle types. Performance should remain stable when many particles are present, with particle effects behaving comparably to their performance in Minecraft 1.12.2 rather than causing the game to become nearly unplayable.

ENVIRONMENT

Minecraft Java Edition Versions: 18w02a, 18w03b, 18w05a, 18w06a, 18w07b, 18w07c, 18w08a, 18w08b, 18w11a, 18w14a, 18w15a, 18w16a, 18w19a, 18w19b, 18w20a, 18w20c, 18w21a, 18w21b, 1.13-pre4, 1.13-pre5, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13-pre9, 1.13-pre10, 1.13, 18w30b, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1-pre2, 1.13.1, 1.13.2-pre1, 1.13.2-pre2, 1.13.2, 18w43b, 18w43c, 18w44a, 18w45a, 18w46a, 18w47a, 18w47b, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w04b, 19w05a, 19w06a, 19w07a, 19w08b, 19w09a, 19w11a, 19w11b, 19w12a, 19w12b, 19w13a, 19w13b, 19w14a, 19w14b, 1.14 Pre-Release 1, 1.14 Pre-Release 2, 1.14 Pre-Release 3, 1.14 Pre-Release 4, 1.14 Pre-Release 5, 1.14, 1.14.1 Pre-Release 1, 1.14.1 Pre-Release 2, 1.14.1, 1.14.2 Pre-Release 1, 1.14.2 Pre-Release 2, 1.14.2 Pre-Release 3, 1.14.2, 1.14.3 Pre-Release 1 (includes snapshots)

ORIGINAL DESCRIPTION

Update:

Also affects my new pc:

CPU: Intel(R) Core(TM) i5-4570 CPU @ 3.20 Ghz

RAM: 8 GB

GPU: GeForce GTX 1050 ti OC Edition

Some fixed in 18w09a, but not all e.g dragon breath and rain particles.

I created a world and I went to the end to defeat enderdragon in survivalmode.

After than the dragon shot a fireball the game is stopped work well, because the area_effect_cloud partice is created many lag spikes.

After I summoned blazes and the fps decreased.

I tried it with every particles

Particle performances in 17w50a-18w11a:

In small render distance(2 chunk), max. fps set 60, played 100 particles with a command(always active repeater command_block)

-angry_villager(100 particle) 33-35 fps

-barrier(100 particle) 33-35 fps

-bubble(100 paricles) 60 fps

-cloud(100 particles) 1-5 fps

-crit(100 particles) 49-52 fps

-damage_indicator(100 particles) 12-26 fps

-dragon_breath(100 particles) 55 fps

-effect(100 particles) 29 fps

enchant(100 particles) 60 fps

and more... Some not laggy and some very laggy

And very laggy some natural generated particles:

dragon_breath,cloud,blockcrack and smoke(blaze and fire)

So the performance with particles was way better in 1.12.2 than 1.13.2 and 1.14 snapshots.

Code analysis

See 1.13 Particles.png

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-124170>